

Ryan Cho

Senior Data Engineer | Cloud Data Platforms

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PROFESSIONAL SUMMARY

Data engineer with six years of experience designing ELT/ETL pipelines, Snowflake warehouses, and real-time streaming systems in regulated financial environments. Skilled in Apache Spark, AWS Glue, Airflow, Kafka, Oracle, and MongoDB for end-to-end ingestion, modeling, and self-service reporting. Provides production support for pipelines and datasets while partnering with analysts and business stakeholders to improve data access.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and deployed **ELT pipelines** on **AWS Glue** and **Apache Spark**, processing 12M+ daily records from trading, risk, and client systems into a centralized **Snowflake** warehouse.
- Built **Apache Airflow** orchestration for 40+ scheduled workflows, reducing pipeline failures by 35% through retry policies, SLA monitoring, and dependency-aware DAG design.
- Designed **cloud-native data architectures** in Snowflake with layered staging, curated, and consumption schemas supporting analytics and regulatory reporting.
- Implemented **real-time streaming ingestion** with **Apache Kafka** and Spark Structured Streaming for market data and portfolio event feeds with sub-minute latency.
- Developed **Oracle** and **MongoDB** data models normalizing legacy holdings data and document-style client profiles for unified downstream consumption.
- Published **self-service reporting assets** and semantic views enabling analysts to explore datasets without engineering dependencies, cutting ad-hoc request volume by 28%.
- Led **incident response and operational support** for warehouse jobs and datasets, maintaining 99.5% pipeline availability through runbooks and on-call rotation.
- Mentored two engineers on **data pipeline standards**, code review practices, and safe deployment patterns for production ingestion workflows.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Built **ETL pipelines** using **Apache Spark** and internal batch frameworks to consolidate trade, credit, and compliance data across global banking divisions.
- Deployed **Airflow-managed workflows** migrating nightly batch jobs to orchestrated DAGs with improved lineage tracking and failure notifications.
- Supported **Snowflake migration** initiatives by mapping legacy **Oracle** schemas and validating row-level reconciliation across source systems.
- Modeled relational and document datasets in Oracle and **MongoDB** supporting risk dashboards and regulatory submissions in financial services.
- Integrated **Kafka streams** for intraday position and transaction events feeding near-real-time risk exposure calculations.
- Partnered with **data consumers** and business analysts to refine ingestion specs, improving dataset freshness and field-level documentation.
- Provided **production troubleshooting** for warehouse loads and pipeline incidents, restoring SLAs within defined escalation windows.
- Automated **data quality checks** and reconciliation scripts reducing manual validation effort across monthly close processes.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Developed **Python data ingestion scripts** pulling vendor API data into PostgreSQL tables for inventory and compliance tracking.
- Assisted senior engineers in building **batch ETL jobs** transforming raw CSV exports into normalized schemas for internal dashboards.
- Wrote **SQL queries** and lightweight reports supporting operations teams reviewing daily reconciliation outputs.
- Contributed to **Airflow DAG** prototypes for scheduled data refreshes and monitoring failed task alerts.
- Documented **pipeline configurations** and data dictionary entries to improve handoff between engineering and product teams.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Built **responsive web interfaces** consuming REST APIs for nonprofit program data visualization and donor reporting.
- Implemented **client-side data fetching** and caching patterns reducing page load times for analytics views.
- Collaborated with backend developers to align **JSON payloads** with UI display requirements for reporting modules.
- Resolved **cross-browser rendering** issues and accessibility gaps across public-facing content pages.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML5, CSS3, D3.js, Chart.js

Backend: Python, Java, Scala, Apache Spark, PySpark, REST APIs, Node.js

Databases: Snowflake, Oracle, MongoDB, PostgreSQL, Redis, SQL, NoSQL modeling

Cloud & DevOps: AWS (Glue, S3, Lambda, IAM), Docker, Terraform, CI/CD, Git, Linux

Monitoring & Tools: Apache Airflow, Apache Kafka, AWS Glue, Datadog, Grafana, Splunk, dbt

Workflow & PM: Agile, Scrum, Jira, Confluence, stakeholder communication, incident management, technical documentation